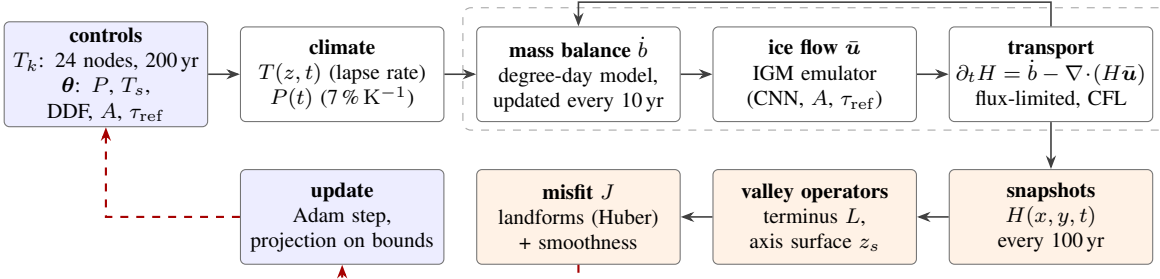
next time step, $\approx 8 \times 10^4$ steps for 4.6 kyr
controls
T_k: 24 nodes, 200 yr
$\boldsymbol{\theta}$: P, T_s,
DDF, A, τ_{ref}
climate
$T(z,t)$ (lapse rate)
$P(t)$ (7 % K^{-1})
mass balance $\dot{b}$
degree-day model,
updated every 10 yr
ice flow $\bar{\boldsymbol{u}}$
IGM emulator
(CNN, A, τ_{ref})
transport
$\partial_t H = \dot{b} - \nabla\cdot(H\bar{\boldsymbol{u}})$
flux-limited, CFL
snapshots
$H(x,y,t)$
every 100 yr
valley operators
terminus L,
axis surface z_s
misfit J
landforms (Huber)
+ smoothness
update
Adam step,
projection on bounds
gradient $\partial J/\partial T_k$, $\partial J/\partial \boldsymbol{\theta}$: backward pass by automatic differentiation through all time steps, with checkpointing (Sect. 2.5)